

Evaluation of lead aprons and their maintenance and management at our hospital

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Abstract Lead aprons are worn by medical workers to reduce the effects of the radiation doses to which they are exposed during radiography and surgery performed with radiosopic apparatus. Regarding the management of such aprons, the Radiation Protection Section of the Japanese Society of Radiological Technology issued the “Guidelines for the Management of Lead Aprons” in 2000, and common management criteria have been set for all institutions. However, we found that the lead aprons used in operating rooms had not been closely inspected before 2014 in our hospital. Thus, we examined the extent of damage of such aprons in our operation room via computed tomography (CT) scout imaging, as well as visual and tactile inspections. Although no abnormality was detected upon visual and tactile inspections, CT images revealed that protective aprons used for 6 years or more had damaged internal radiation shields, thus risking radiation exposure. In response to these results, we fully realized the need to examine the date of the initial use of currently used lead aprons, to routinely perform visual and tactile inspections, and to regularly evaluate the extent of damage to the internal radiation shields via fluoroscopy in cooperation with the radiation management section.

Keywords Lead apron · Internal radiation shields · Radiation management

Introduction

When surgery is performed with a radiosopic apparatus, as in osteosynthesis in orthopedic surgery and stent placement in cardiovascular surgery, medical workers wear lead aprons to prevent their radiation exposure. In Japan, regarding the methods for routine management of these lead aprons, the Radiation Protection Section of the Japanese Society of Radiological Technology issued the “Guidelines for the Management of Lead Aprons” in 2000 [1]. The commonly used lead aprons usually have a radiation shield that is composed of a rubber or plastic sheet containing high ratios of shielding elements (lead and other elements), and the surface of the shield is covered with a protective sheet. At our hospital, we found in 2014 that only visual inspections had been performed for routine management of the lead aprons in the past, and that the condition of the internal radiation shields had not been closely inspected. Thus, although we again performed visual and tactile inspections of the lead aprons used in operating rooms in our hospital, we also inspected the condition of the internal radiation shields with a computed tomography (CT) device in cooperation with the radiation management section.

The specimens examined in the study included 35 lead aprons used in operating rooms. The control numbers assigned to the lead aprons were checked, and the duration of usage after their initial use was determined. In the visual inspection, the state and extent of damage to the protective sheets on the surface of the lead aprons were inspected. In the tactile inspection, the state of damage to the internal radiation shields was inspected by touching the lead aprons with the fingertips. To determine the state of damage to the internal radiation shields of the lead aprons without destroying them, internal damage was inspected via the

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Table 1 Years since initial use of lead aprons examined in the study with the stage classification

Stage	Number of aprons	Initial use (years ago)				
		14	9	6	2	0
1	14	0	0	0	6	8
2	1	0	0	1	0	0
3	6	0	4	2	0	0
4	5	0	5	0	0	0
5	9	3	3	3	0	0

$n = 35$

Stage 1 no damaged area, *Stage 2* one damaged area, *Stage 3* two or more damaged areas but no problem with further use at this point, *Stage 4* two or more damaged areas (further use is not advisable), *Stage 5* many damaged areas make the aprons unsuitable for further use

acquisition of computed tomography (CT) scout images obtained with a CT diagnostic device (SOMATOM Definition Edge; Siemens, München, Germany). Because a single lead apron could not be entirely captured in a single image with the use of an ordinary radioscopic device, we used CT scout images instead. Based on images obtained by this procedure, the extent of damage to the lead apron was classified into five stages: stage 1, no damaged area; stage 2, one damaged area; stage 3, two damaged areas or more but no problem with further use at this time; stage 4, two damaged areas or more, and further use is not advisable; and stage 5, many damaged areas make the aprons unsuitable for further use.

Table 1 lists the protective aprons used according to the duration after their initial use. Duration varied widely, ranging from 0 to 14 years. Three aprons had been used for the longest duration, 14 years. Although visual inspection revealed certain aprons had fraying hook-and-loop fasteners and so forth, we found no apron with its protective sheet damaged sufficiently to allow exposure of the internal radiation shield. Moreover, tactile inspection also revealed no apparent abnormal finding. From radioscopic CT images, we determined nine aprons to be too far damaged to be used further, and all these aprons had been used for 14, 9, or 6 years. We also confirmed that the extent of damage varied even among the protective aprons used for the same duration.

In our inspection of lead aprons that had been used for 6 years or more, evaluation of CT images revealed that the internal radiation shields unexpectedly had many damaged areas (Fig. 1; stage 5, a–f). Moreover, among those that were still usable, some had damaged areas that appeared prone to further deterioration (Fig. 1; stage 3). In all lead aprons including those having extensively damaged internal radiation shields as revealed by imaging tests, no abnormality of the external protective sheet was detected upon conventional visual inspection.

In comparison to a normal lead apron (Fig. 1; stage 1), a possible cause of the observed damage to the hem and

shoulder parts of the internal radiation shield is deterioration over time as a result of the weight of the protective apron itself. The damage to the waist part can be attributed to the belt that is fastened to reduce the load on the shoulders while wearing the lead apron. Moreover, the extent of damage may be affected by failure to correctly follow daily storage procedures, such as leaving the lead aprons carelessly folded for a long period of time instead of hanging them on special hangers.

Regarding the historical development of lead aprons, those developed in Japan in the 1950s were made by sewing lead-containing rubber materials. These protective aprons were heavy and extremely inflexible; they were extremely unpopular among medical workers. Subsequently, the shape of radiation shields was improved, and materials that are lighter and more flexible than rubber were used. At present, flexible and lead-free protective aprons are predominantly used [2]. It is difficult to detect damage in the radiation shields made of such flexible thin materials by inspecting their appearance alone, and it appears to be difficult to detect abnormalities even by tactile inspection.

In Japan, the Japanese Society of Radiological Technology issued the “Guidelines for the Management of Lead Aprons” in 2000, specifying the proper use, storage, and quality control of lead aprons [1]. However, because the usage of the aprons varies across institutions, the guidelines only encourage each institution to prepare management manuals appropriate to their actual usage conditions on the basis of the guidelines.

Their aprons should be monitored in terms of safety and protection, and periodic control is also required, so their internal structure should be tested regularly, starting from the time of purchase, at least once every year [3]. In addition, the manufacturers of many generally distributed lead aprons recommend the following methods of inspection and testing: visual inspection (the hem bulges when the radiation shield drops off), tactile inspection, and radioscopy performed every 6 months (aprons manufactured 3 years ago or earlier should be tested more frequently).

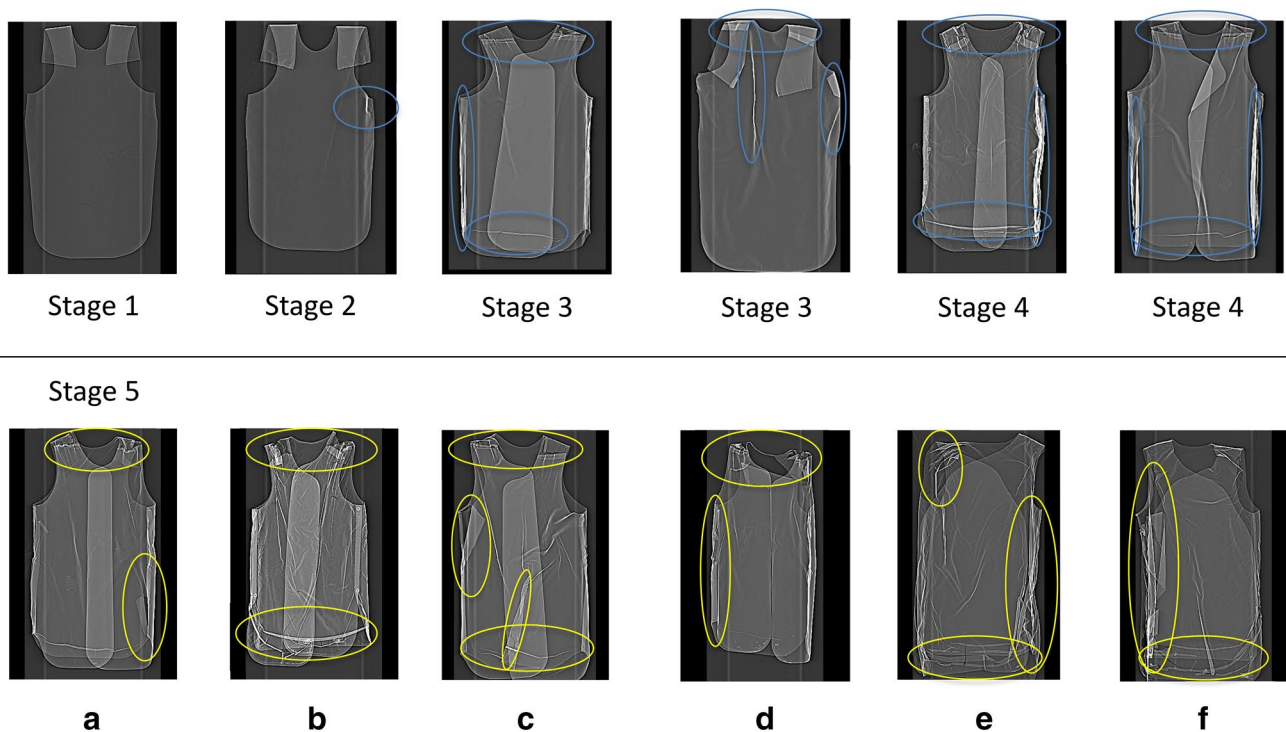


Fig. 1 Upper: Examples of lead aprons of stages 1, 2, 3, and 4: a normal lead apron (stage 1) and lead aprons of stages 2, 3, and 4 (as per our classification). Blue circles indicate areas of damage. Lower: Example of a lead apron of stage 5 (as per our classification) that was determined to be unsuitable for further use. Yellow circles indicate damaged areas. Possible cause of the observed damage to the hem

and shoulder parts of the internal radiation shield is deterioration over time caused by the weight of the protective apron itself. Damage to the waist area can be attributed to the belt that when fastened reduces the load on the shoulders while wearing the lead apron. a, b Used for 14 years; c, d used for 9 years; e, f used for 6 years

Although the average estimated usable period is 5 years, as a matter of course, this is not applicable to all cases because the usage varies at each institution.

Regarding the actual acceptable levels of damage to the radiation shields, those with a hole with a diameter of up to 7 mm or a slit of up to 7 cm can be used further, whereas damage exceeding these levels is considered to induce the risk of radiation exposure to the inside of the apron [4]. Lead aprons that were determined unusable (Fig. 1a–f) in this study were all consistent with this criterion. So, in such a case, it is desirable to discontinue the use of the lead apron. In the past, there was also a case in which a physician performing cardiac catheterization was exposed to radiation because of the delayed detection of great damage to the radiation shield of his lead apron [5].

In addition, there are various sizes of lead aprons for different body types. When a person of small stature wears a large-size lead apron, the spaces around the chest and the armpits increase, causing a problem with protection against scattered radiation. Moreover, because sitting and other body movements are likely to cause creases in the radiation shield, there is also a concern about the resultant impact on the durability of the aprons. In this study, finding anomalies

by visual and tactile inspections was not possible. However, with inspection by visual observation, there is a possibility of finding the exposure of the radiation shields by damage by fraying to a suture or a protective sheet. Moreover, when carefully applying tactile inspection the lesion site found by CT scout imaging, it was possible to sense the internal abnormalities. However, in recent years sensing of abnormalities in the thin internal radiation shield seems quite difficult, in contrast to the thicker old type of shielding in which detecting the abnormalities may be possible.

We plan to utilize the findings in this study to recommend immediate inspection of routinely used lead aprons if inspection systems for them are flawed.

In conclusion, this study revealed that the following measures are necessary to guarantee that medical workers do not suffer from unnecessary radiation exposure and to ensure complete management of lead aprons in future. It is important to know the year of the initial use of currently used lead aprons and to perform routine visual and tactile inspections and regular X-ray examinations such as CT scout imaging in close cooperation with a radiation management section. Moreover, it is also necessary to ensure proper storage. For example, lead aprons need to be stored

correctly in the designated place immediately after their use. Furthermore, it may be preferable for medical workers to routinely carry a film badge for complete personal monitoring and to wear a lead apron of a size appropriate to their body type.

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